

# A Controlled Study of Pure Monte Carlo Tree Search for Symbolic Regression

After conversations with GPT

August 20, 2026

## 1 Setup

### 1.1 Problem

Data  $(x_i, y_i)_{i=1}^n$  are generated by a fixed unknown expression  $h$ ,  $y_i = h(x_i)$ . Symbolic regression searches a discrete set  $\Pi$  of expression *structures*; a structure  $\pi \in \Pi$  carries free coefficients  $\beta$ , fitted by least squares,

$$\hat{\beta}_\pi = \arg \min_{\beta} \frac{1}{n} \sum_{i=1}^n (y_i - \pi_\beta(x_i))^2,$$

and is scored by the coefficient of determination of that fit, clipped to  $[-1, 1]$  and written  $R(\pi)$  (6). The clip is necessary, not cosmetic: fitting  $\beta x$  to `lin_C` of Table 2 returns  $R^2 \approx -2.9 \times 10^{12}$ , and one such completion would otherwise decide the mean value of a whole subtree. The inner fit is part of the score, so  $R$  is a function of the structure alone and the symbolic search is  $\pi^* = \arg \max_{\pi \in \Pi} R(\pi)$ . The motivating instance is a quadratic target  $h(x) = c_0 + c_1x + c_2x^2$ , with structures ranging over the subsets of  $\{1, x, x^2\}$ .

The structure search becomes *subset selection*. The purpose of the study is to characterize the finite-budget limitations of pure Monte Carlo tree search (MCTS) in grammar-based symbolic regression. The neural-guidance question is scientifically separate: before asking whether a network improves search, one must determine what information the underlying pure-search procedure estimates and when that information is aligned with symbolic optimization.

### 1.2 Values and failure modes

A state  $s$  is a partial derivation,  $\mathcal{A}(s)$  its set of legal actions, and  $T : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$  the deterministic transition. Let

$$\mathcal{E}(s) = \{\pi : \pi \text{ is a terminal expression completable from } s\},$$

so that  $\Pi = \mathcal{E}(s_0)$  is the structure set of §1 and a terminal expression  $\pi$  is itself a state, with  $\mathcal{E}(\pi) = \{\pi\}$ . The best-achievable value is

$$V^*(s) = \max_{\pi \in \mathcal{E}(s)} R(\pi), \tag{1}$$

where  $R(\pi)$  is the terminal symbolic-regression reward. The uniform completion policy

$$q(a \mid s) = \frac{1}{|\mathcal{A}(s)|}, \quad a \in \mathcal{A}(s),$$

induces, because transitions are deterministic, a distribution  $q(\cdot \mid s) \in \Delta(\mathcal{E}(s))$  over completions; the mean-rollout value is

$$V^q(s) = \mathbb{E}_{\pi \sim q(\cdot \mid s)} [R(\pi)]. \tag{2}$$

Every target below is exactly expressible in the grammar, so  $V^*(s_0) = 1$ , and the exact-completion mass is

$$\rho(s) = \Pr_{\pi \sim q(\cdot|s)} [R(\pi) \geq 1 - \tau], \quad \tau = 10^{-6}, \quad (3)$$

the probability that a uniform completion of  $s$  fits the target exactly. The threshold is the *global* optimum  $V^*(s_0) = 1$ , not the local  $V^*(s)$ , so it is the same at every state and  $\rho$  obeys the same backward induction as  $V^q$ . A single tolerance  $\tau$  serves here and in the exact-recovery indicator (23); on the six targets that carry the study every exact fit returns  $R = 1$  to the last bit, so no reported value of  $\rho$  depends on it (Remark 3.2).

**Question 1.1.** Which mechanisms make pure MCTS fail at finite budget?

Two candidate mechanisms:

1. *Value-semantic failure*: for sibling states  $s_1, s_2$ ,

$$V^q(s_1) > V^q(s_2) \quad \text{while} \quad V^*(s_1) < V^*(s_2).$$

Mean-based search then confidently exploits the wrong branch: a shallow decoy yields a high expected reward, while the branch containing the true optimum is penalized by its abundance of poor completions.

2. *Rare-event failure*:  $\rho(s) \ll B^{-1}$  at budget  $B$ . Search may explore the correct branch yet exhaust the budget before sampling a near-optimal leaf.

Therefore  $V^q(s)$ ,  $V^*(s)$  and  $\rho(s)$  are the three primary metrics of every experiment below. Table 1 summarizes the two mechanisms.

| Failure mechanism      | Formal condition                                                                | Intuitive cause                                                                       | Implication for search                                                                                  |
|------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Value-semantic failure | $V^q(s_1) > V^q(s_2)$ while $V^*(s_1) < V^*(s_2)$ for sibling states $s_1, s_2$ | A suboptimal branch has the higher mean rollout reward: a shallow decoy looks better  | Search confidently exploits the wrong branch and starves the branch containing the true global optimum. |
| Rare-event failure     | $\rho(s) \ll B^{-1}$                                                            | Near-optimal completions carry a negligible fraction of the rollout probability mass. | Search explores the correct branch but exhausts the budget $B$ before sampling a near-optimal leaf.     |

Table 1: The two finite-budget failure mechanisms of mean-based MCTS.

**Remark 1.2.** The optimization objective depends on  $V^*$ , whereas conventional UCT uses empirical mean returns as its exploitation statistic [6]; asymptotic consistency does not imply effective finite-budget search, and adverse finite-depth behavior is known for UCT [3]. The goal in symbolic regression is one exceptional expression, not a policy with high average return: best-focused Monte Carlo methods [1, 2], risk-seeking symbolic regression [8], and recent MCTS systems with learned proposals or extreme-value allocation [4, 5] all respond to this mismatch.

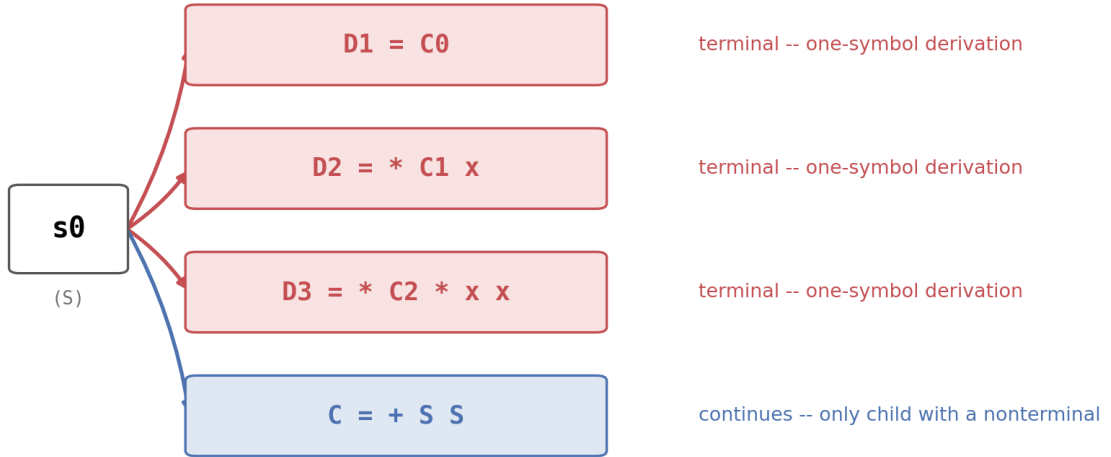
Experiment 1 measures these quantities exactly on a controlled family. Experiment 2 tests whether they predict the finite-budget behavior of pure MCTS.

### 1.3 Notation

Three families of symbol are easy to confuse and are kept apart throughout: math italic  $c_p$  for the coefficients of the *target*, typewriter C0, C1, C2 for the *grammar tokens* whose values are fitted, and  $C$  for a distinguished *state*.

| Symbol                                 | Type                                                         | Meaning                                                                       |
|----------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------------|
| $s \in \mathcal{S}$                    | state                                                        | sentential form in the length- $L$ buffer                                     |
| $s_0$                                  | state                                                        | root, the one-symbol form ( $S$ )                                             |
| $\pi \in \Pi$                          | state                                                        | terminal expression; $\Pi = \mathcal{E}(s_0)$                                 |
| $\mathcal{E}(s)$                       | set of states                                                | terminals completable from $s$                                                |
| $a = \text{pos} \cdot P + \text{prod}$ | action                                                       | flattened (pos, prod) index                                                   |
| $\mathcal{A}(s)$                       | set of actions                                               | the legal mask at $s$ , (5)                                                   |
| $T(s, a)$                              | map $\mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$ | deterministic transition                                                      |
| $w_0, \dots, w_{P-1}$                  | symbol strings                                               | the productions, in code order                                                |
| $S, A$                                 | nonterminals of $G$                                          | italic; calligraphic $\mathcal{S}, \mathcal{A}$ are the state and action sets |
| $R(\pi)$                               | $\mathbb{R}$                                                 | $\text{clip}(R^2, -1, 1)$ after the inner fit, (6)                            |
| $V^*, V^q, \rho$                       | $\mathcal{S} \rightarrow \mathbb{R}$                         | (1), (2), (3)                                                                 |
| $C, D_1, D_2, D_3$                     | states                                                       | the four children of $s_0$ , Section 3.1                                      |
| $c_0, c_1, c_2$                        | $\mathbb{R}$                                                 | the target's chosen coefficients, (7)                                         |
| C0,C1,C2                               | grammar tokens                                               | free constants fitted by <code>lmfit</code>                                   |
| $B, K$                                 | $\mathbb{N}$                                                 | terminal-evaluation budget; independent completions                           |
| $\tau$                                 | $\mathbb{R}$                                                 | exactness tolerance $10^{-6}$ , (3)                                           |

root  $s_0$ : 4 legal actions, one per production, 3 terminal + 1 that continues



from ADDITIVE\_GRAMMAR via `compile_grammar` -- no hand-picked numbers

The reachable states form a directed acyclic graph rather than a tree: an action names a buffer position, so expanding the two nonterminals of  $+ S S$  in either order reaches the same form. Experiment 1 therefore runs a backward induction over 4,898 *states*, whereas the search tree Experiment 2 builds is a tree over *action sequences*, in which one form may occupy several nodes.

## 2 The derivation MDP in the code

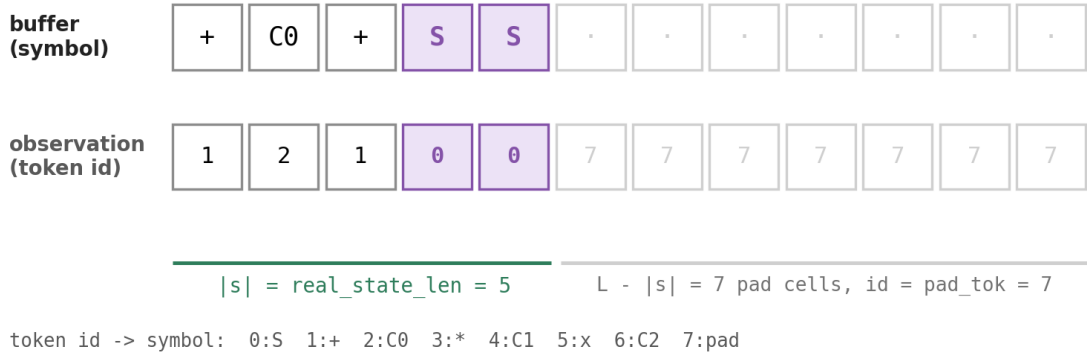
**Definition 2.1** (The grammar-derivation MDP). Fix a context-free grammar  $G$  with start symbol  $S$ , production set  $\{w_0, \dots, w_{P-1}\}$  indexed in code order, and a buffer length  $L$  (`max_len`). The environment is the deterministic undiscounted MDP:

- *State*. A sentential form  $s \in \Sigma^*$  with  $|s| \leq L$ , stored as a fixed-length array of  $L$  symbols together with  $|s| = \text{real\_state\_len}$ ; cells beyond  $|s|$  hold a pad symbol. The observation space is

$$\text{MultiDiscrete}([\lceil |\Sigma| + 1 \rceil]^L)$$

, so the pad symbol is observable and the agent sees how much buffer remains.<sup>1</sup>

`L = 12 fixed-length buffer` | `observation_space = MultiDiscrete([8] * 12)`

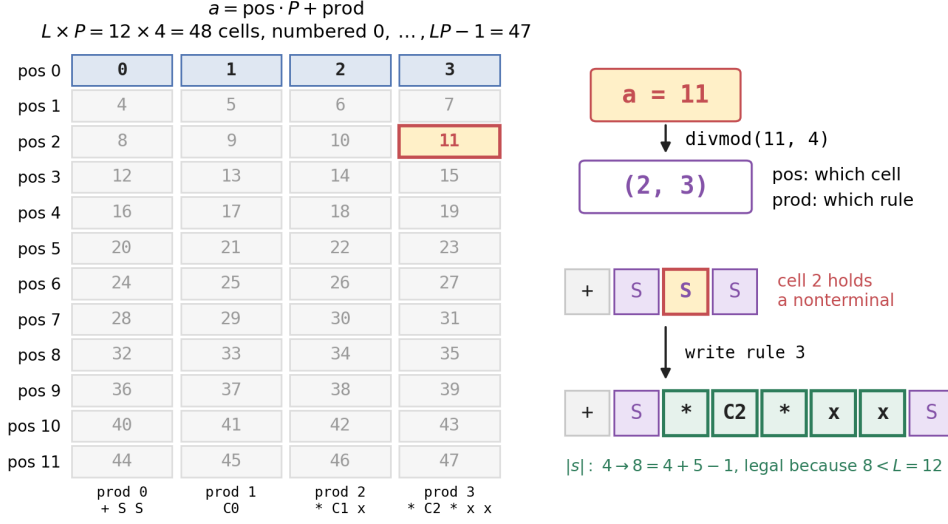


- *Start*.  $s_0 = (S)$ , a single nonterminal,  $|s_0| = 1$ .
- *Action*. An action must say *which* cell of the buffer to rewrite and *which* production to write there, so the action set is the set of pairs  $(\text{pos}, \text{prod}) \in \{0, \dots, L-1\} \times \{0, \dots, P-1\}$ . There are  $LP$  such pairs, and `Discrete(L*P)` indexes them by one integer counted from zero, whence

$$a \in \{0, \dots, LP - 1\}, \quad a = \text{pos} \cdot P + \text{prod}, \quad (\text{pos}, \text{prod}) = \text{divmod}(a, P). \quad (4)$$

The  $-1$  is zero-indexing of  $LP$  cells and nothing more; at  $L = 12$ ,  $P = 4$  the largest action is 47.

<sup>1</sup>Necessary, not cosmetic: the policy/value net is a fixed-width MLP over a one-hot encoding of the whole buffer,  $L(|\Sigma| + 1)$  inputs built by `SymRegPolicyValueNet._encode`. `real_state_len` is never itself passed to the network, only `state` is, so a pad symbol in the vocabulary is the only way a fixed-shape input still carries  $|s|$ : the network recovers it by counting non-pad slots.



shaded row: the only cells that exist at  $s_0$ , where  $|s_0| = 1$ ; masking, not the index set, decides which actions are legal

- *Legal set.*  $a \in \mathcal{A}(s)$  iff

$$\text{pos} < |s|, \quad s[\text{pos}] \text{ is a nonterminal}, \quad \text{prod} \in \text{proddict}[s[\text{pos}]], \quad |s| + |w_{\text{prod}}| - 1 < L. \quad (5)$$

- *Transition.*  $T(s, a)$  splices  $w_{\text{prod}}$  over cell  $\text{pos}$ , shifts the tail right, and re-pads. It is a function, not a kernel.
- *Reward.*  $R \equiv 0$  at every nonterminal state. At a terminal, the symbol string is read as a prefix expression, its  $\mathcal{C}^*$  tokens are fitted by `lmfit`, and the reward is the clipped coefficient of determination

$$R(\pi) = \text{clip}(R^2, -1, 1), \quad (6)$$

with  $-1$  returned for any unparseable expression or failed solve.

- *Discount.*  $\gamma = 1$ ; exactly one reward per episode is nonzero, so the return of an episode is its terminal  $R^2$ .

(a) the MDP as implemented in `sraz/instances/symreg/game.py` — every value below is read off the running game

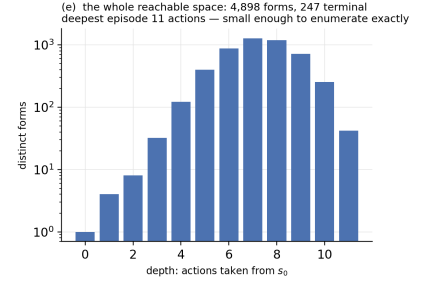
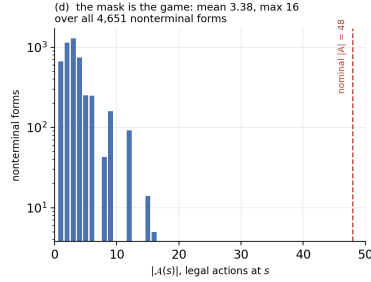
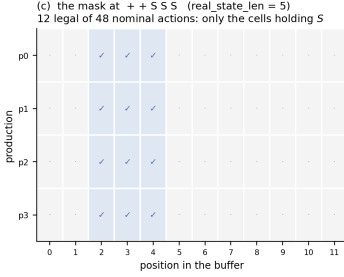
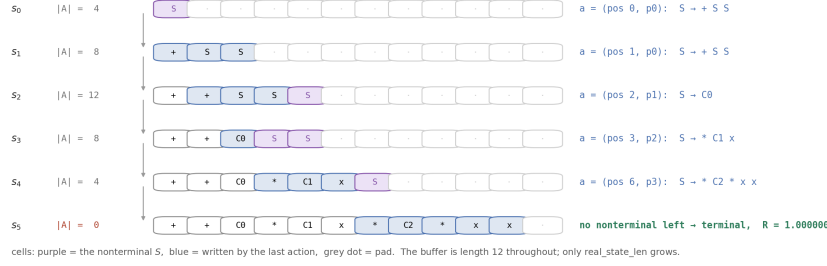
```

state s      token buffer of fixed length 12, plus real_state_len; observation space MultiDiscrete([8] * 12)
start s0     [S] then 11 pad — a single nonterminal, real_state_len = 1
action a      Discrete(48) = state_len × nprods = 12 × 4; decoded (pos, prod) = divmod(a, 4)
legal set ℒ(s) pos < real_state_len, buffer[pos] is a nonterminal, and len(s) + |rhs| − 1 < state_len
transition T   deterministic: splice the production's rhs over buffer[pos], shift the tail right, re-pad
reward R       0 at every nonterminal state; at a terminal, clip(R2 of the lmfit constant fit, −1, 1)

the four productions, one nonterminal S
p0: S → + S S      |rhs| = 3      Δlen = +2
p1: S → C0         |rhs| = 1      Δlen = +0      escape production: never lengthens the buffer, so it is never masked out
p2: S → * C1 x      |rhs| = 3      Δlen = +2
p3: S → * C2 * x x   |rhs| = 5      Δlen = +4
tokens ['S', '+', 'C0', '*', 'C1', 'x', 'C2'] + pad(7); the target is C0 + C1*x + C2*x**2; γ = 1, and the only nonzero reward is terminal.

```

(b) one episode, replayed from the game: the action (pos, prod) rewrites one nonterminal cell into the production's right-hand side



Two features of (4)–(5) matter for everything downstream. First, the action space is nominally *LP* but almost entirely illegal: on the additive instance at  $L = 12$ ,  $LP = 48$  while the legal set averages 3.38 and never exceeds 16 over all 4,651 nonterminal forms. The mask, not the action space, is what search branches on. Second, because an action names a position, the uniform completion policy  $q$  of (2) is uniform over the flattened (pos, prod) mask, *not* uniform over the productions of one designated nonterminal. The two measures differ as soon as a form holds more than one nonterminal, and  $V^q$  inherits whichever is used, so which one is meant must be stated.

## 2.1 Termination

The environment has no step limit, no truncation, and no stop action. An episode ends when the sentential form runs out of nonterminals, and not before. Concretely, `step` has three exits, and only the first can fire under masked play.

**Exit 1: normal termination.** `terminated = ¬_has_nonterms(s)`. The finished string is decoded, converted to infix, fitted, and scored by (6). This is how every episode in every experiment ends.

**Exit 2: invalid action.** A defensive branch returns  $R = -1$  with `terminated = true` when the decoded action is out of range, addresses a terminal cell, or would overflow the buffer. Calling `step` off-mask on  $s_0$  reproduces it exactly. Under masked play it is unreachable; it exists so that a masking bug fails loudly rather than looping.

**Exit 3: dead end.** A form retains a nonterminal but  $\mathcal{A}(s) = \emptyset$ . Then **step** is never called at all: `MCTS._rollout_value` breaks out of its loop and *discards* the sample rather than scoring it.

[Proposition 2.3](#) shows this exit cannot fire.

**Proposition 2.2** (Length bound). Every reachable form satisfies  $|s| \leq L - 1$ . In particular every terminal expression has at most  $L - 1$  symbols.

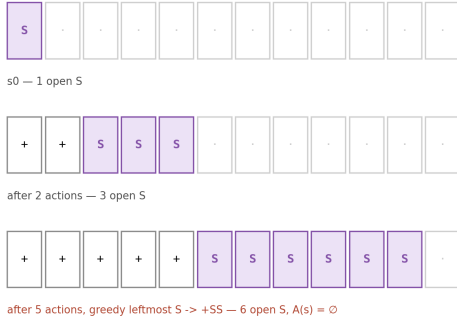
The bound is attained, and the strict inequality in (5) matters: at  $L = 12$  the longest terminals have 11 symbols, and the exact expression  $+ C0 + * C1 x * C2 * x x$  is admissible so that  $V^*(s_0) = 1$ . At  $L = 11$  it is excluded, and  $V^*(s_0)$  then depends on the target: it stays at 1 for every linear target, which needs only five symbols, but falls to 0.997213 on `quad_B` and 0.527856 on `quad_A` ([Section 3](#)). A buffer must be set one longer than the longest expression one intends to be reachable.

**Proposition 2.3** (No dead ends). Suppose every nonterminal  $A$  of  $G$  admits a production  $A \rightarrow w$  with  $|w| = 1$ , an *escape production*. Then  $\mathcal{A}(s) \neq \emptyset$  for every reachable  $s$  containing a nonterminal, so Exit 3 never fires.

*Proof.* Let  $s[i] = A$  be a nonterminal and  $A \rightarrow w$  an escape production. The first three conditions of (5) hold at that pair by construction. For the fourth,  $|s| + |w| - 1 = |s| \leq L - 1 < L$  by [Proposition 2.2](#). The action is legal, so the mask is nonempty.  $\square$

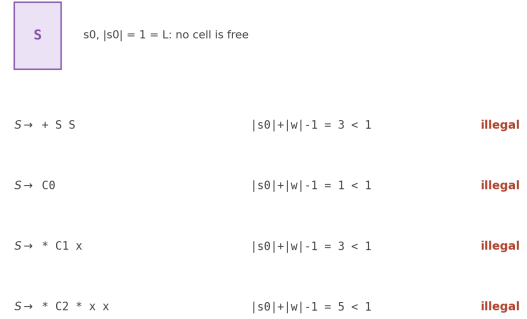
Both shipped grammars satisfy the hypothesis through  $S \rightarrow C0$ ; an exhaustive census finds 0 dead ends among the 4,898 reachable forms. The hypothesis is a genuine constraint rather than a formality. Removing the constant production, or capping  $L$  below the shortest completion of some reachable form, reintroduces dead ends — and since the rollout discards them instead of scoring them, they would bias  $V^q$  silently rather than raise an error.

(a) no escape production: drop  $S \rightarrow C0$  from the grammar



every remaining production has  $|w| \geq 3$ : at length 11,  $|s| + |w| - 1 \geq 13$ , not  $< 12 = L$ , for any of them. 6 nonterminals are stuck at once; a rollout through here discards the sample instead of scoring it.

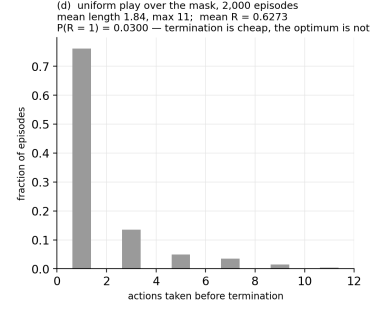
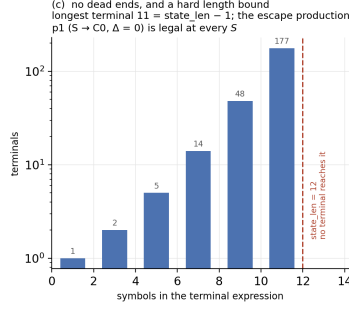
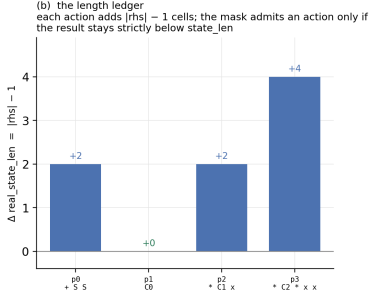
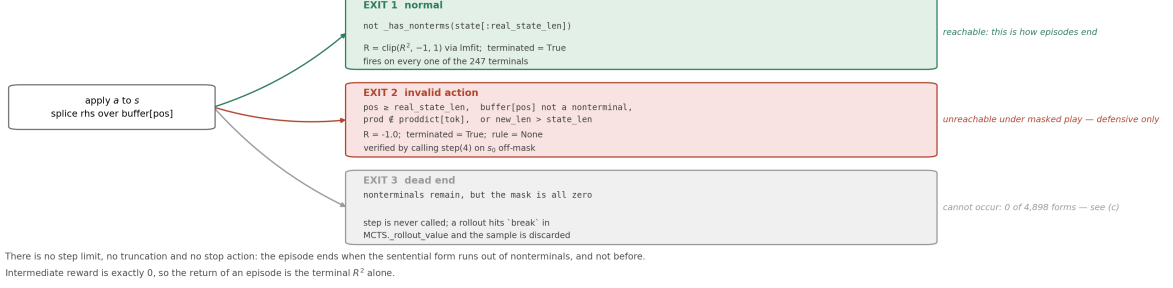
(b) buffer too small: the real grammar at  $L = 1$



even the zero-growth escape production needs  $|s_0| < L$ ; at  $L = 1$  it fails like every other row.  $\mathcal{A}(s_0) = \emptyset$  while  $s_0$  still holds a nonterminal: the root itself is a dead end.

**Remark 2.4** (Parity). Each action changes the length by  $|w| - 1 \in \{0, 2, 4\}$ , all even, and  $|s_0| = 1$ . Every reachable form therefore has odd length, and terminals can only hold 1, 3, 5, 7, 9, 11 symbols. The census agrees exactly: 1, 2, 5, 14, 48, 177 terminals at those six lengths. [Appendix A](#) lines up four reachable terminals by length.

(a) termination: the three exits in `SymRegGame.step`, and which of them can actually fire



### 3 A minimal controlled family

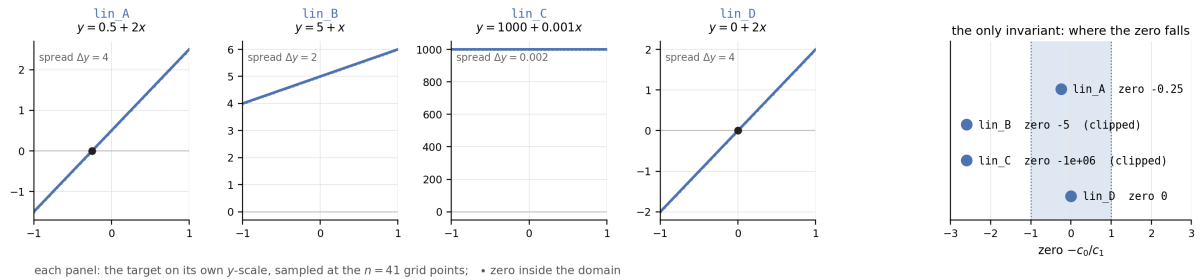
The MDP of Definition 2.1 is now held fixed — grammar `ADDITIVE_GRAMMAR`, buffer  $L = 12$  — and the target the terminal expression is fitted against is the only thing that varies.

**Definition 3.1** (Target family). A *target* is a polynomial together with the grid it is sampled on,

$$y(x) = \sum_{p=0}^d c_p x^p, \quad x_1, \dots, x_n \text{ equally spaced in } [-1, 1], \quad n = 41, \quad (7)$$

with  $d \leq 2$  and coefficients *chosen*. The family has eight members: four linear, separated by where the zero  $-c_0/c_1$  falls relative to the domain, and four quadratic, separated by the vertex  $x_v = -c_1/2c_2$  and the discriminant  $\text{disc} = c_1^2 - 4c_0c_2$ .<sup>2</sup>

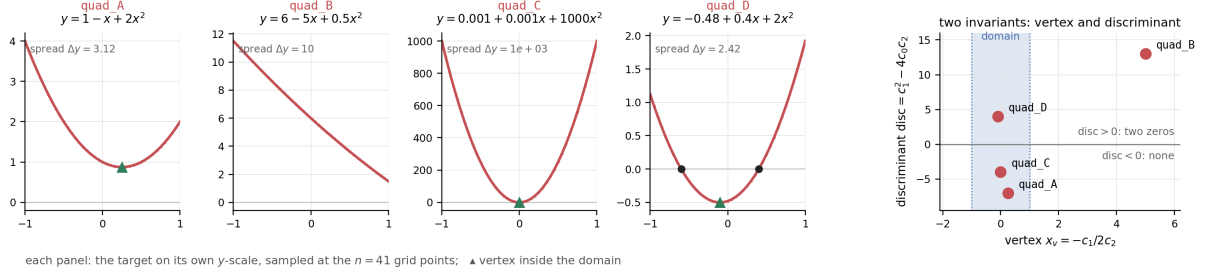
The four linear targets, separated by where the zero falls:



The four quadratic targets, separated by the vertex and the discriminant:

<sup>2</sup> $x_v$  says where the curve turns and  $\text{disc}$  whether it meets the axis at all ( $\text{disc} > 0$  two real zeros,  $< 0$  none), so the pair fixes the shape on  $[-1, 1]$ : of the four members only **quad\_B** is monotone there, and only **quad\_D** crosses zero inside the domain. The word *root* is reserved throughout for the MDP root  $s_0$ .





### 3.1 Where the difficulty is

**Key hypothesis.** Two mechanisms, and only two, are hypothesized to drive finite-budget failure of mean-based search: value-semantic failure at the root, which starves the branch containing the optimum, and rarity of the exact completion, which imposes a sample-efficiency floor no matter how search is allocated (Section 1.2). The two are dissociable by construction: replacing the mean backup with a maximum backup (Section 5) should cure the first and leave the second untouched. What follows locates, on the eight-target family of Definition 3.1, exactly where each mechanism appears and which targets isolate it — formalized as Hypotheses 2a–2c in Section 5.

Exactly  $|\mathcal{A}(s_0)| = 4$  actions are legal at the root, one per production, and they have four distinct successors. We name those four successor *states* after the production that creates them,

$$D_1 = C0, \quad D_2 = * C1 \ x, \quad D_3 = * C2 \ * \ x \ x, \quad C = + \ S \ S, \quad (8)$$

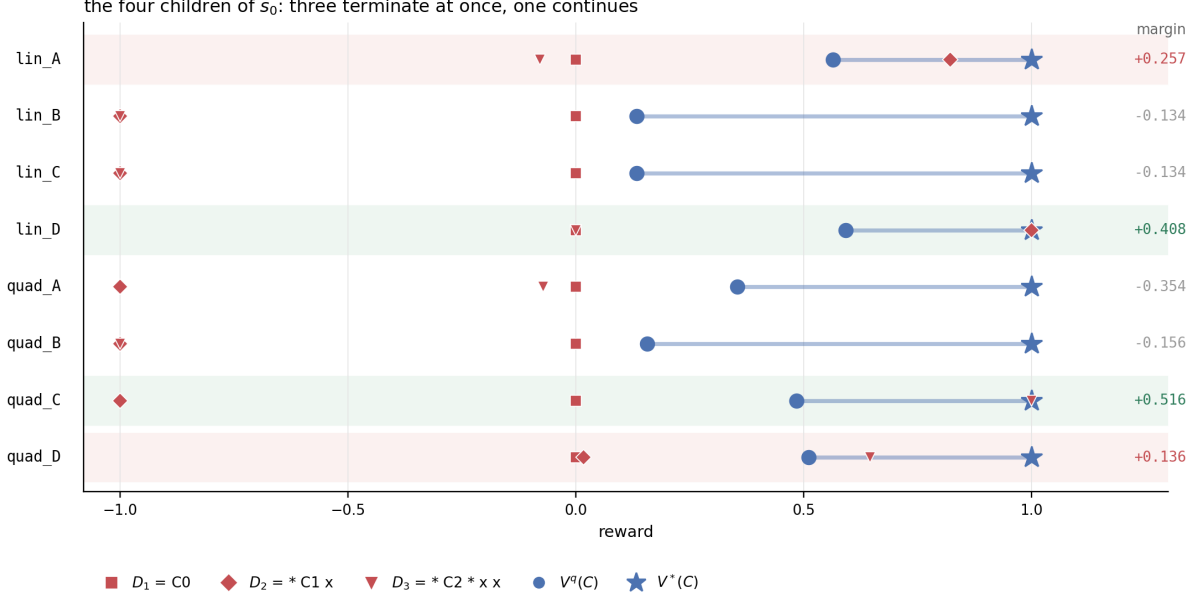
the same four states already previewed in Section 1.3. Because the four root actions are in bijection with these four children, we also write  $N(s_0, C')$  for the visit count of the edge  $s_0 \rightarrow C'$ ; no separate symbol for a root action is needed.

Three of the four productions carry no nonterminal on the right, so  $D_1$ ,  $D_2$  and  $D_3$  are already terminal: each is a one-action terminal with  $V^*(D_i) = V^q(D_i) = R(D_i)$ , one deterministic number. Each also fits exactly one monomial —  $D_1$  the constant 1,  $D_2$  the linear term  $x$ ,  $D_3$  the quadratic term  $x^2$  — and we call this monomial the *atom* of  $D_i$ , indexed by its degree  $p \in \{0, 1, 2\}$ ; a general terminal is a sum of the atoms its derivation happens to include. The fourth child  $C$  is the only one that continues, and the only one whose subtree contains an exact expression:  $V^*(C) = 1$  for every target. Root value-semantic failure is therefore a single inequality,

$$\max_i R(D_i) > V^q(C) \quad \text{while} \quad \max_i R(D_i) < V^*(C) = 1, \quad (9)$$

and nothing has to be added to the environment to produce it. We call (9) an *inversion*, and a target on which it holds a *trap*: the one-action terminal that mean rollout prefers is not the one whose subtree contains the optimum.

The continue branch’s  $[V^q(C), V^*(C)]$  interval against the three one-action terminals, per target:



**Remark 3.2** (Conditioning). `lin_C` and `quad_C` deliberately span a coefficient ratio of  $10^6$  (Definition 3.1). The inner fit is an iterative nonlinear solver (`lmfit`) even though every production here is linear in its constants, so on these two targets it terminates short of the closed-form optimum: the best terminal reaches  $R = 1 - 8.7 \times 10^{-13}$  on `lin_C` and  $R = 1 - 8.1 \times 10^{-12}$  on `quad_C`, against bit-exact 1 on the other six targets. Both residuals sit far inside the tolerance  $\tau = 10^{-6}$  of (3), so no reported value of  $\rho$  or  $R$  changes, but the exactness is true to floating-point precision rather than to the last bit.

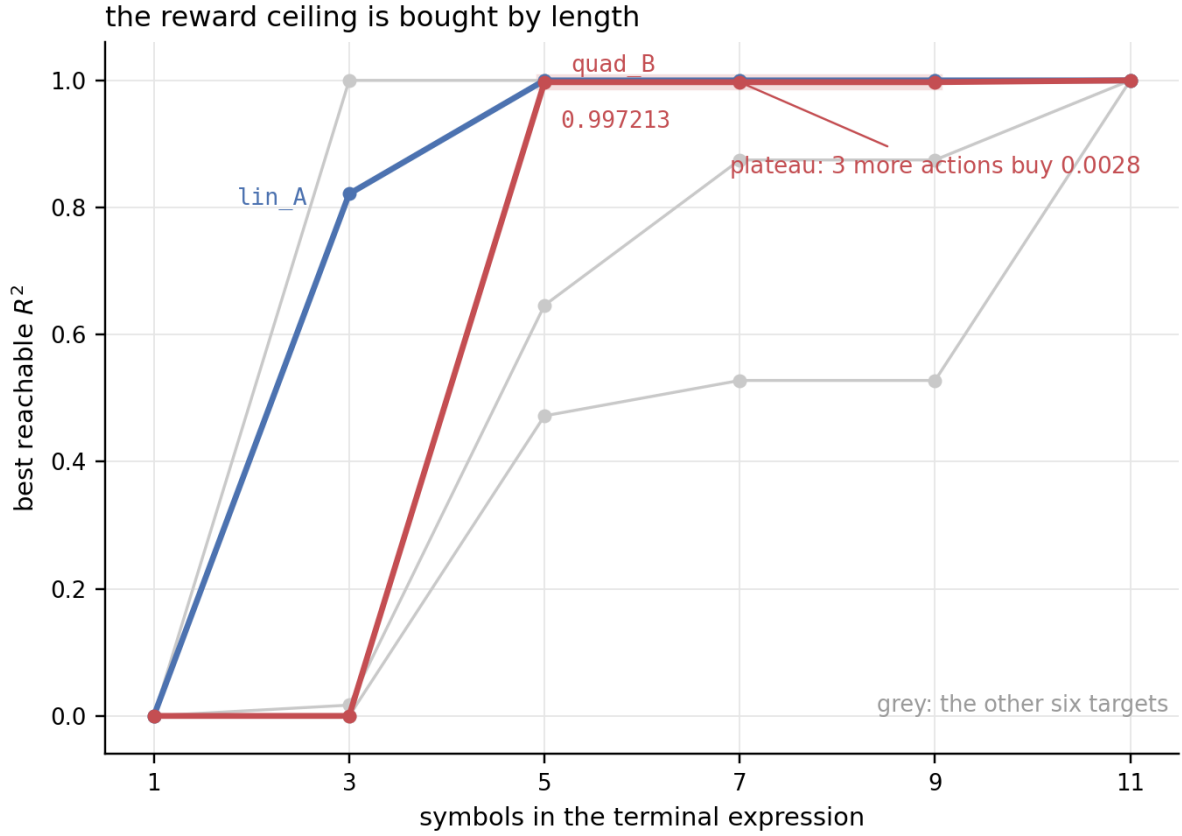
Both sides of (9) are tabulated in Table 2. Only `lin_A` and `quad_D` satisfy it. For `lin_D` and `quad_C` the margin  $\max_i R(D_i) - V^q(C)$  is also positive, but there the one-action terminal is itself exact — for `quad_C` only to within round-off, Remark 3.2 — so preferring it is correct and no trap exists. The second inequality of (9), not the margin alone, separates the two cases.

| Target | $y(x)$                     | $R(D_1)$ | $R(D_2)$ | $R(D_3)$ | $V^q(C)$ | $\rho(C)$ | margin  |                |
|--------|----------------------------|----------|----------|----------|----------|-----------|---------|----------------|
| lin_A  | $0.5 + 2x$                 | 0        | 0.8214   | -0.0793  | 0.5646   | 0.3945    | +0.2569 | trap           |
| lin_B  | $5 + x$                    | 0        | -1       | -1       | 0.1341   | 0.3945    | -0.1341 | —              |
| lin_C  | $1000 + 0.001x$            | 0        | -1       | -1       | 0.1341   | 0.3945    | -0.1341 | —              |
| lin_D  | $2x$                       | 0        | 1        | 0        | 0.5924   | 0.5924    | +0.4076 | solved at once |
| quad_A | $1 - x + 2x^2$             | 0        | -1       | -0.0711  | 0.3540   | 0.1128    | -0.3540 | —              |
| quad_B | $6 - 5x + 0.5x^2$          | 0        | -1       | -1       | 0.1563   | 0.1128    | -0.1563 | plateau        |
| quad_C | $0.001 + 0.001x + 1000x^2$ | 0        | -1       | 1        | 0.4840   | 0.5569    | +0.5160 | solved at once |
| quad_D | $-0.48 + 0.4x + 2x^2$      | 0        | 0.0170   | 0.6461   | 0.5106   | 0.1128    | +0.1355 | trap           |

Table 2: Exact root quantities for the eight targets under one fixed MDP;  $V^*(C) = 1$  in every row. “Trap” marks the rows satisfying (9). The  $\rho(C)$  column is at  $\tau = 10^{-6}$ ; it is tolerance-free except on `lin_C` and `quad_C` (Remark 3.2).

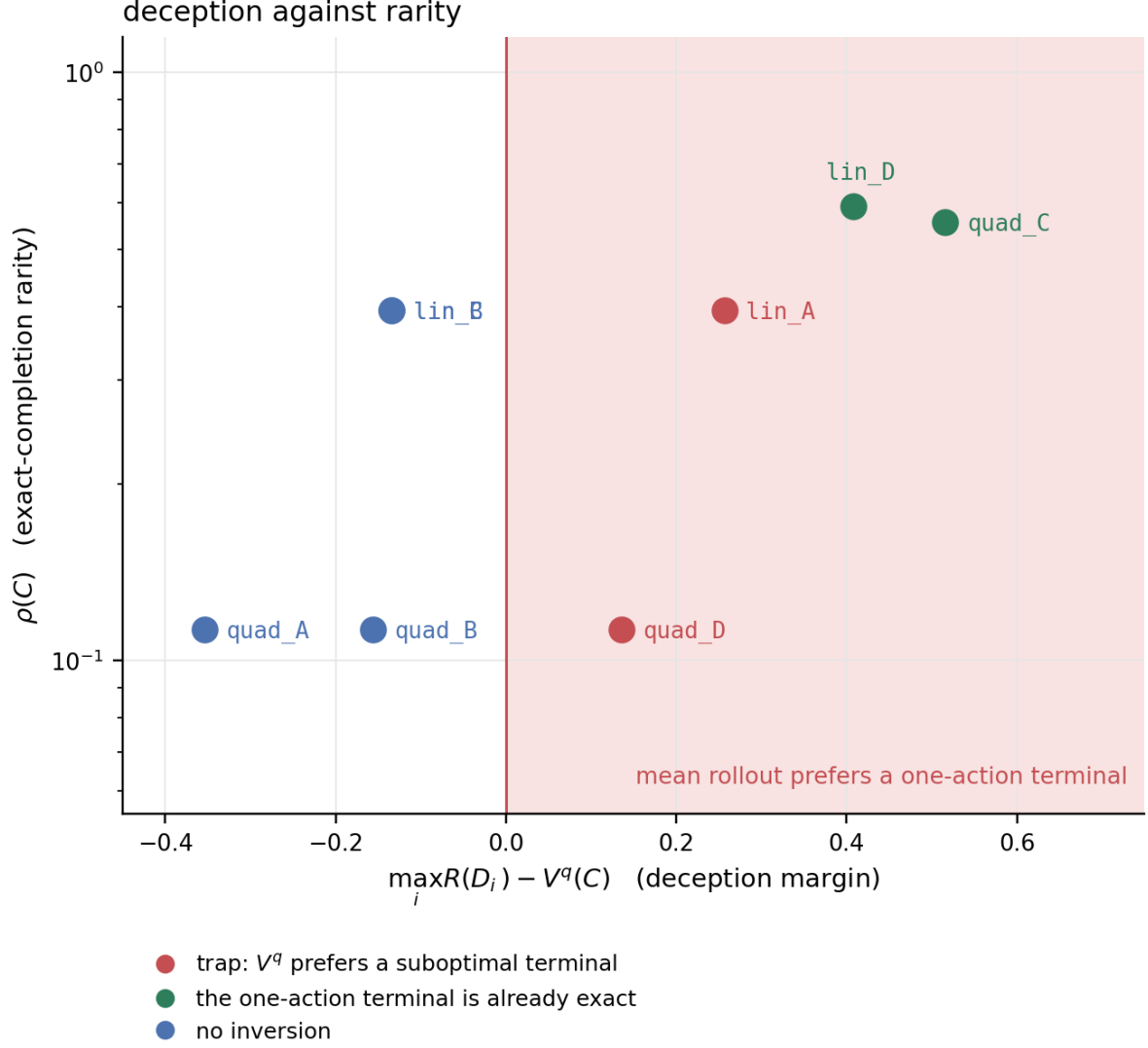
For the linear family,  $R(D_1), R(D_2), R(D_3)$  above also have a closed form in  $c_0, c_1$  (Appendix B).

The reward ceiling as a function of terminal length, per target:



The reward ceiling is the best  $R^2$  any terminal of  $\ell$  symbols attains. For `quad_B` the ceiling is 0.997213 already at  $\ell = 5$ , stays there at  $\ell = 7$  and  $\ell = 9$ , and reaches 1 only at  $\ell = 11$ . Three of the five actions of an optimal derivation buy 0.0028 of  $R^2$  between them. The trap of (9) reappears one level down as a plateau inside the continue branch, not a decoy action at the root: a fitted *line* through a parabola whose vertex sits at  $x_v = 5$ , far outside the domain, already explains 99.7% of the variance.

**Rarity is a property of the target's support.** The tree is the same for every target, so the only thing that can vary is which of its terminals fit.  
Deception margin against rarity, per target:



Each point is one target, plotted by the same two quantities as (9): the deception margin  $\max_i R(D_i) - V^q(C)$  on the  $x$ -axis, and the rarity  $\rho(C)$  on the  $y$ -axis, log-scaled because it spans an order of magnitude. Rarity is  $\rho$  of (3) evaluated at  $C$ : the probability that a uniform random completion of the continue branch fits the target exactly. A low point is therefore one where an exact fit is a rare event among random completions of  $C$ , regardless of whether that branch is also a trap. Color marks the root outcome: red for a trap (the shaded region,  $\max_i R(D_i) > V^q(C)$ ), green where the one-action terminal is already exact, blue where the ordering is correct and untrapped.

**Lemma 3.3** (Rarity depends only on the support). Let  $P = \{p : c_p \neq 0\}$  be the support of the target (7), and  $A(\pi) \subseteq \{0, 1, 2\}$  the set of atom degrees a terminal  $\pi$  carries. Then  $\pi$  fits the target exactly iff  $A(\pi) \supseteq P$ . Hence  $\rho$  depends on the target only through  $P$ , and

$$\rho(C) = 0.5924, \quad 0.3945, \quad 0.1128 \quad \text{for} \quad P = \{1\}, \quad \{0, 1\}, \quad \{0, 1, 2\}.$$

*Proof.* The atoms  $1, x, x^2$  are linearly independent on the 41-point grid, so a terminal's fitted value lies in the span of the atoms in  $A(\pi)$ . If  $A(\pi) \supseteq P$  that span contains the target, and least squares is exact: any atoms beyond  $P$  do not prevent this, since their coefficients are simply fit to zero. If instead some  $p \in P$  is missing from  $A(\pi)$ , linear independence gives the target a genuinely nonzero component along that atom that no combination of the others can reproduce, so the fit is strictly inexact. The predicate  $R(\pi) \geq 1 - \tau$  therefore depends on  $\pi$  only

through  $A(\pi)$ , and  $\rho$ , a uniform average of that predicate over completions (16), inherits the dependence.  $\square$

Of the 247 terminals, 146 carry both 1 and  $x$  while only 12 carry all three, so the family offers exactly two rarity levels. Since completions are independent,  $K$  of them miss with probability  $(1 - \rho)^K$ , so  $\Pr(\text{at least one exact}) \geq 1 - \delta$  exactly when

$$K \geq K_{1-\delta} = \left\lceil \frac{\log \delta}{\log(1 - \rho)} \right\rceil, \quad (10)$$

which is 6 completions at  $\rho = 0.39$  and 26 at  $\rho = 0.11$ , for  $\delta = 0.05$ . Here  $K$  counts completions drawn from one fixed state, whereas the budget  $B$  of (20) counts terminal evaluations anywhere in the tree.  $K$  is what *uniform* sampling from  $C$  needs; a tree search is bound by it in neither direction, and Section 5 measures the difference.

## 4 Experiment 1: exact value-semantic audit

**Question 4.1.** Does the mean random-completion return  $V^q$  preserve the ordering induced by the best-achievable reward  $V^*$ , and how rare ( $\rho$ ) is the exact expression within each partial derivation?

**Hypothesis 1.** The rollout mean ranks the continue branch above every one-action terminal on `lin_B` and `quad_A`, and below one of them on `lin_A` and `quad_D`:

$$\begin{aligned} V^q(C) &> \max_i R(D_i) && \text{on lin\_B and quad\_A,} \\ V^q(C) &< \max_i R(D_i) && \text{on lin\_A and quad\_D,} \end{aligned} \quad (11)$$

with  $V^*(C) = 1 > \max_i R(D_i)$  throughout, so the second inequality is a genuine ordering failure. Independently, the completion budget required for exact recovery scales as

$$K_{1-\delta} = \Theta\left(\frac{\log(1/\delta)}{\rho}\right) \quad \text{as } \rho \rightarrow 0. \quad (12)$$

**Procedure.** No stochastic tree search is used in this experiment. Enumerate the mask-reachable forms once — 4,898 of them, 247 terminal — and for each of the four cells compute exactly, at every state  $s$ ,

$$V^*(s), \quad V^q(s), \quad \rho(s). \quad (13)$$

For a nonterminal state  $s$ , define the action selected by the mean-rollout oracle as

$$a_q(s) = \arg \max_a V^q(T(s, a)), \quad (14)$$

and define its oracle sibling regret by

$$g_q(s) = V^*(s) - V^*(T(s, a_q(s))). \quad (15)$$

The reachable set is a directed acyclic graph — every action either lengthens the form or removes a nonterminal — so all three quantities follow by one backward induction over it,

$$V^*(s) = \max_a V^*(T(s, a)), \quad V^q(s) = \frac{1}{|\mathcal{A}(s)|} \sum_{a \in \mathcal{A}(s)} V^q(T(s, a)), \quad (16)$$

and likewise for  $\rho$ , with terminal states seeded by their reward. The fitter is called once per terminal and cached, so a cell costs 247 fits, not 4,898.

Two checks guard the census. The reward of the exact derivation + C0 + \* C1 x \* C2 \* x must come back as 1 to fit tolerance in every cell, and the induced  $V^q(s_0)$  must agree with the mean of  $10^5$  uniform masked rollouts to within Monte Carlo error. Failure of either is an implementation error, not a finding.

**Primary outputs.** Report the exact root quantities of Table 2 —  $R(D_i)$ ,  $V^*(C)$ ,  $V^q(C)$ ,  $\rho(C)$ ,  $K_{0.95}$  — and plot:

1. the terminal reward distribution over the 247 terminals, per cell;
2.  $V^q(s)$  against  $V^*(s)$ , stratified by  $|s|$ ;
3. sibling regret  $g_q(s)$  against  $|s|$ ;
4. the exact hit curve  $1 - (1 - \rho)^K$ .

Because Experiment 1 is an exact census, it requires no random seeds, confidence intervals, or significance tests.

## 5 Experiment 2: finite-budget pure-MCTS limits

**Question.** At matched terminal-evaluation budgets, do the value-order mismatch and exact-completion rarity identified in Experiment 1 predict the behavior of actual pure search?

### 5.1 Search conditions

We compare three conditions.

**Uniform random search.** Sample complete derivations independently from the uniform grammar policy and return the best terminal expression observed.

**Mean-UCT.** At node  $s$ , select

$$a_t = \arg \max_a \left[ \bar{Q}_t(s, a) + \sqrt{\frac{2 \log N_t(s)}{N_t(s, a)}} \right], \quad (17)$$

where  $N_t(s)$  and  $N_t(s, a)$  count the simulations through  $s$  and through  $(s, a)$  after  $t$  simulations, and  $\bar{Q}_t(s, a)$  is the empirical mean of terminal returns from simulations passing through  $(s, a)$ . Unvisited actions receive infinite score. Rewards lie in  $[-1, 1]$ , so (17) applies the UCB1 bonus for a unit-range reward to a range-2 one; the constant is held fixed across all conditions rather than tuned, so it cannot by itself explain a difference between them. Each simulation selects through the current tree, expands one unvisited edge, performs one uniform random completion, and backs up the terminal reward by an arithmetic mean.

**Max-UCT diagnostic.** Define

$$Q_t^{\max}(s, a) = \max \{R(\pi_b) : \pi_b \text{ was sampled through } (s, a)\}. \quad (18)$$

Select using

$$a_t = \arg \max_a \left[ Q_t^{\max}(s, a) + \sqrt{\frac{2 \log N_t(s)}{N_t(s, a)}} \right]. \quad (19)$$

The tree structure, expansion policy, rollout policy, and exploration term are otherwise identical to Mean-UCT. This condition is a causal diagnostic of the mean-versus-maximum distinction, not a claim that naive maximum backup is universally optimal. A calibrated extreme-bandit allocation rule is a natural subsequent comparison [4].

## 5.2 Evaluation protocol

Each search run uses one persistent tree rooted at  $s_0$ . The algorithm does not commit to a grammar action before the budget is exhausted. No network, learned prior, Dirichlet noise, reward clipping, complexity penalty, or nonlinear optimizer is used. One simulation contains exactly one terminal reward evaluation.

Use the terminal-evaluation budgets

$$\mathcal{B} = \{16, 32, 64, 128, 256, 512, 1024, 2048, 4096, 8192\}. \quad (20)$$

For each of the four benchmark cells and each search condition, use 100 independent seeds. Run each seed once to  $B_{\max} = 8192$  and record all intermediate checkpoints. The same base-seed list is used across algorithms.

## 5.3 Outcomes

Let  $\pi_1, \dots, \pi_B$  be the terminal expressions observed by budget  $B$ . Define the best-observed reward

$$R_B^{\max} = \max_{1 \leq b \leq B} R(\pi_b), \quad (21)$$

the simple regret

$$r_B = 1 - R_B^{\max}, \quad (22)$$

and the exact-recovery indicator

$$Z_B = \mathbf{1} \{R_B^{\max} \geq 1 - \tau\}. \quad (23)$$

The budget at which exact recovery first occurs is

$$B_{\text{exact}} = \inf\{B : Z_B = 1\}, \quad (24)$$

written with  $Z$  and  $B$  rather than  $S$  and  $T$ , which [Section 1.3](#) reserves for the start symbol and the transition map.

Exact recovery is the primary outcome. Simple regret is secondary because a structurally wrong expression already achieves small regret: on `quad_B` the plateau of [Section 3.1](#) sits at

$$1 - 0.997213 = 0.002787,$$

and on `lin_A` the decoy  $D_2 = * \text{ C1 x}$  sits at  $1 - 0.821429 = 0.178571$  without containing a constant term at all. Thus predictive reward alone cannot distinguish near-perfect approximation from structural recovery. This distinction is consistent with symbolic regression benchmarks that report both predictive and exact-recovery performance [7].

For mechanism diagnosis, record the continue-visit fraction

$$F_B = \frac{N_B(s_0, C)}{N_B(s_0)}, \quad (25)$$

where  $N_B(s_0)$  and  $N_B(s_0, C)$  are the root and continue-edge visit counts at budget  $B$ , together with the final root recommendation, number of distinct terminal sequences, number of distinct canonical atom sets, and the best reward found through each root action. The final visit-count recommendation is diagnostic only; the primary search output is the best terminal expression encountered.

## 5.4 Hypotheses

**Hypothesis 2a: value-semantic effect.** Mean-UCT will allocate less search to the continue branch on the inversion cells (`lin_A`, `quad_D`) than on their non-inversion partners (`lin_B`, `quad_A`), because its exploitation statistic is anchored near  $V^q(C)$  while subtree sampling remains close to uniform, and  $V^q(C)$  lies below  $\max_i R(D_i)$  exactly on those cells. Max-UCT will be less sensitive to the inversion after observing high-reward continuations.

**Hypothesis 2b: rarity effect.** For every algorithm, exact recovery will be slower on the quadratic cells ( $\rho(C) = 0.11$ ) than on the linear ones (0.39). At moderate budgets, recovery will be organized approximately by the effective rare-event budget

$$B\rho. \quad (26)$$

**Hypothesis 2c: interaction.** The largest recovery difference between Max-UCT and Mean-UCT will occur on `quad_D`, the inversion-and-rare cell, where the continue branch has both a lower rollout mean than a one-action terminal and the small exact mass  $\rho(C) = 0.1128$ .

One pre-specified interaction contrast is

$$\begin{aligned} \Delta_{\text{semantic}}(B) = & [P_B^{\text{Max}} - P_B^{\text{Mean}}]_{\text{inversion}} \\ & - [P_B^{\text{Max}} - P_B^{\text{Mean}}]_{\text{no inversion}}, \end{aligned} \quad (27)$$

where  $P_B^{\text{Max}}$  and  $P_B^{\text{Mean}}$  are the exact-recovery probabilities  $\Pr(Z_B = 1)$  of Max-UCT and Mean-UCT, each bracket averaged over the two cells at that level of the design. The prediction is

$$\Delta_{\text{semantic}}(B) > 0,$$

with a larger effect expected on the quadratic (rare) cells.

## 5.5 Statistical reporting and interpretation

For each budget, cell, and algorithm, report the exact-recovery proportion with a 95% Wilson interval, median simple regret with a seed-bootstrap interval, the empirical survival curve

$$\Pr(B_{\text{exact}} > B),$$

and the continue-visit fraction  $F_B$ .

The interpretation is mechanism-based:

- If Mean-UCT fails while Max-UCT succeeds, the mean-return statistic is causally implicated.
- If both methods deteriorate primarily on the quadratic cells, completion rarity is the dominant limitation.
- If random search outperforms Mean-UCT on an inversion cell, the tree allocation rule actively suppresses the optimal branch.
- If Mean-UCT succeeds despite the exact value-order inversion, UCT exploration is sufficient at the tested budgets and the mismatch alone is not a complete explanation.
- If Max-UCT performs worse on the non-inversion cells, maximum backup is overoptimistic and should not be treated as a universal replacement for mean backup.
- If simple regret is small while exact recovery remains poor, the experiment demonstrates a separation between predictive approximation and symbolic identification.



## 5.6 Secondary validation and deferred questions

After the two primary experiments, a nonconfirmatory validation may repeat the key comparison on the full seven-production grammar, whose sine and division productions are pure distractors for these targets: at  $L = 12$  it has 795,437 reachable forms and 19,438 terminals against the 4,898 and 247 used above, so rarity there is a grammar effect rather than a target effect. Nonlinear constant fitting, uniform-prior PUCT, learned policy and value heads, and SRBench evaluation enter only after the controlled mechanisms are understood. In particular, whether a learned policy or value function corrects the pure-search failures identified here — or instead learns and amplifies the biased targets produced by mean-return search — is deferred, so that the present study remains causally interpretable.

## A Symbols in a terminal expression

A *symbol* is one cell of the state buffer of [Definition 2.1](#) — what [Proposition 2.2](#) and [Remark 2.4](#) count. It is not the same as a *term* of the fitted polynomial: written by hand  $c_0 + c_1x$  has two terms, but as a buffer string it is the five-symbol `+ C0 * C1 x`, and the three-term  $c_0 + c_1x + c_2x^2$  is the eleven-symbol string already named in [Section 2.1](#). Below, four reachable terminals lined up by length, symbols on the left and the same expression regrouped on the right; length skips straight from 1 to 3 to 5 because [Remark 2.4](#) makes every even length unreachable.

### symbols in a terminal expression

each row: one reachable terminal, at  $L = 12$

1 symbol

`C0` =  $C_0$

5 symbols

`* C2 * x x` =  $C_2 * x * x$

9 symbols

`+ * C1 x * C2 * x x` =  $C_1 * x + C_2 * x * x$

11 symbols

`+ C0 + * C1 x * C2 * x x` =  $C_0 + C_1 * x + C_2 * x * x$

lengths are always odd -- 1, 3, 5, 7, 9, 11 symbols -- because each application of '+ S S' adds two symbols

## B Closed-form root values for the linear family

For a linear target  $c_0 + c_1x$  ( $c_1 \neq 0$ ), the reward at each one-action terminal of [Section 3.1](#) has a closed form, on the  $n = 41$ -point symmetric grid where  $\sum_i x_i = \sum_i x_i^3 = 0$  and

$$\mu_2 = \frac{1}{n} \sum_i x_i^2 = 0.35, \quad \mu_4 = \frac{1}{n} \sum_i x_i^4 = 0.220325.$$

$D_1$  fits only the constant atom, so its fitted value is the mean  $c_0$  and  $R(D_1) = 0$  always.  $D_2$  fits only the linear atom;  $\sum_i x_i = 0$  makes that fit isolate  $c_1$  exactly and leave the constant entirely unexplained, so

$$R(D_2) = \text{clip}\left(1 - \frac{1}{\mu_2} \left(\frac{c_0}{c_1}\right)^2, -1, 1\right). \quad (28)$$

$D_3$  fits only the quadratic atom;  $\sum_i x_i^3 = 0$  makes it blind to the linear term entirely, so

$$R(D_3) = \text{clip}\left(-\frac{c_0^2}{c_1^2\mu_2}\left(1 - \frac{\mu_2^2}{\mu_4}\right), -1, 1\right) \leq 0, \quad (29)$$

strict whenever  $c_0 \neq 0$ , since  $\mu_4 > \mu_2^2$  by the linear independence of 1 and  $x^2$  on the grid. Both formulas verify exactly against [Table 2](#):  $(c_0/c_1)^2 = 0.0625$  for `lin_A` against  $\mu_2 = 0.35$  keeps (28) at 0.8214, high enough to trap, while  $(c_0/c_1)^2 = 25$  for `lin_B` drives it to  $-1$ .

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